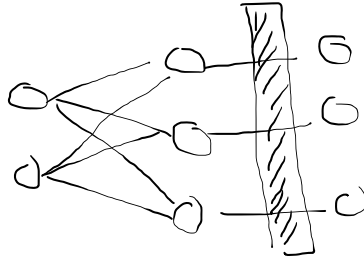


Introduction

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Normalize $\leftarrow 1/P$



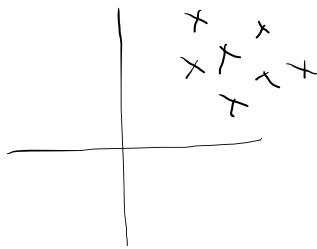
$$Z_{ij} \rightarrow Z_{ij}^{\text{N}}$$

$$\frac{Z_{ij} - \mu}{\sigma}$$

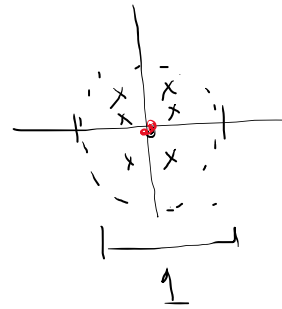
mean

standard deviation

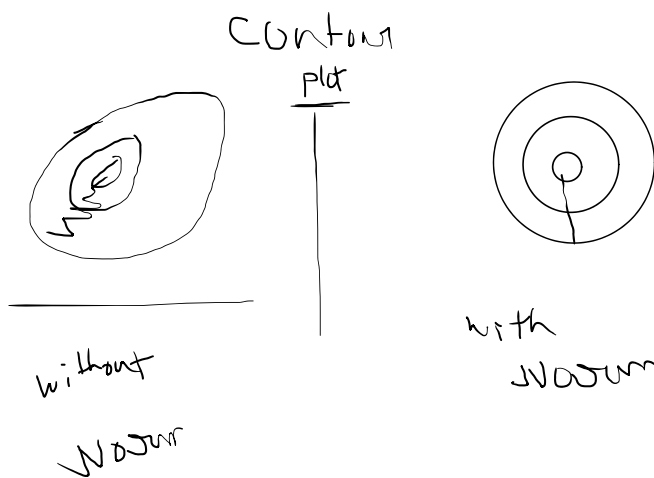
Normalization $\rightarrow \mu = 0$
 $\sigma = 1$



after
Norm



ip $\rightarrow$ [limit]
[Scaling]



Internal Covariate Shift:-

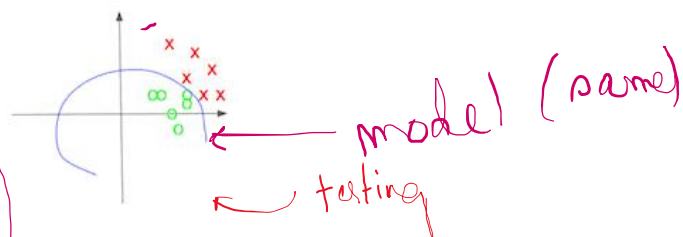
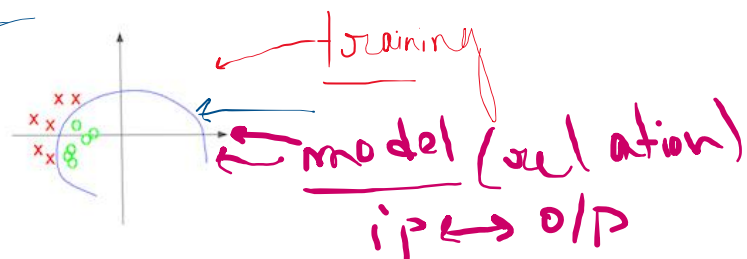
$\rightarrow$ Covariate Shift

Distribution of i/p are diffⁿ in training & testing

Rose (all are red)



✓



Variety of colours

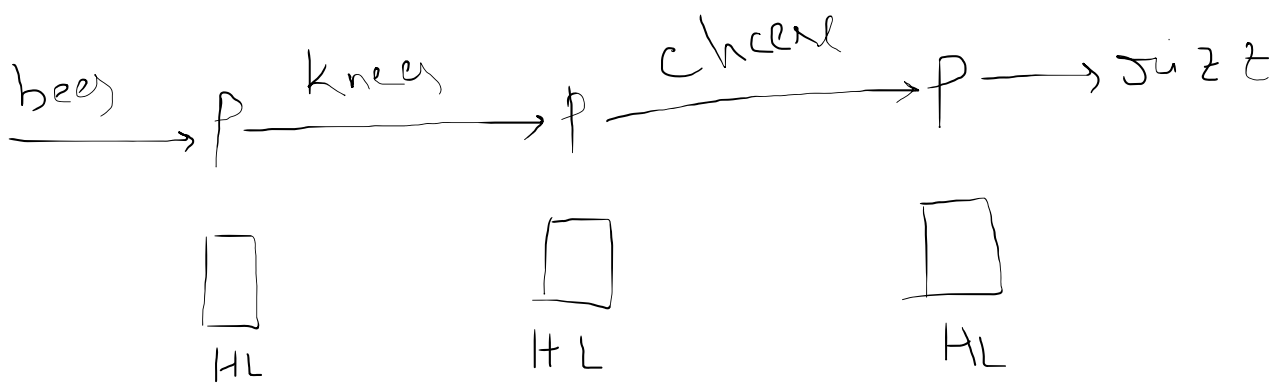
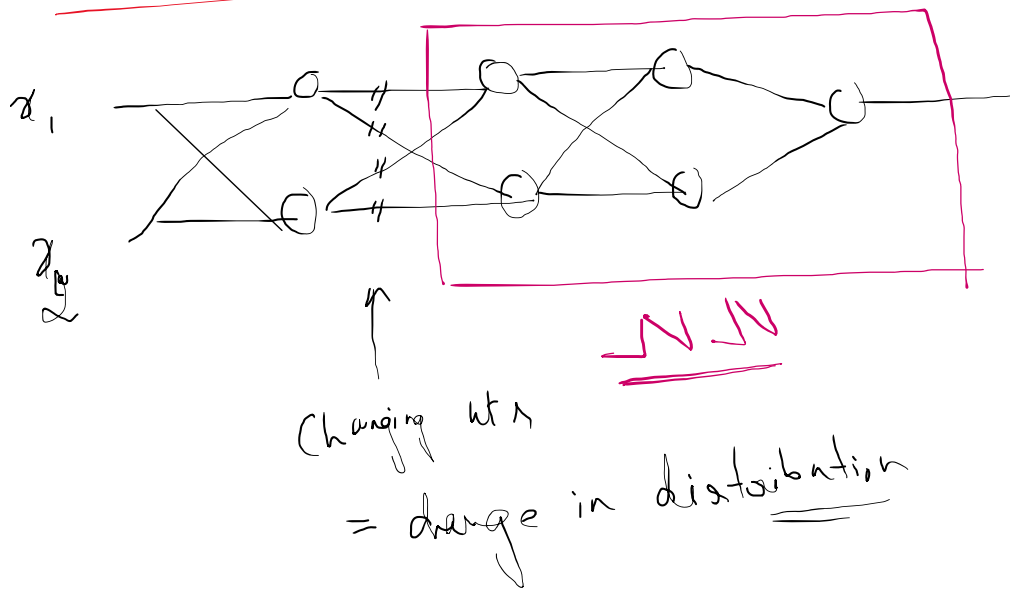
if occurs

Normalize

{ Distribution is diffⁿ in training & testing

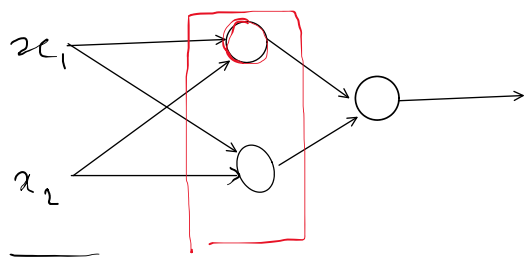
External Covariate Shift:-

→ Internal Covariate Shift:-



← gaussian distribution

Copy	ia	placed
8.9	100	1
6.2	89	0
9.1	91	0
7.7	76	1
...	...	...
6.7	9.1	0



* Mini Batch GD. $\checkmark$
 * layers by layer.

Activation function

$$\textcircled{I} z_{11} \rightarrow z_{11}^{sr} \rightarrow g(z_{11}^{sr}) \rightarrow a_{11}^{sr} \checkmark$$

$$\textcircled{II} z_{11} \rightarrow g(z_{11}) \rightarrow a_{11} \rightarrow a_{11}^{sr}$$

batch=4

$$(4,2) * (2,2) + (1,2)$$

$$= (4,2) + (1,2) = (4,2)$$

$$z_{11}^{sr} = \frac{z_{11} - \mu_B}{\sigma_B} \leftarrow \text{SD}$$

$$\mu_B \checkmark$$

$$\sigma_B \checkmark$$

$$\mu_B = \frac{1}{m} \sum_{i=1}^m z_{11}^i$$

$$\sigma_B = \sqrt{\frac{1}{m} \sum_{i=1}^m (z_{11}^i - \mu_B)^2}$$

point

$$z_{11}^i = \frac{z_{11}^i - \mu_B}{\sigma_B + \epsilon} \leftarrow \text{Epsilon term}$$

$$z_{11}^i = [z_{11}^i + (-\mu_B)] * (\sigma_B + \epsilon)^{-1}$$

$$= (z_{11}^i + \bar{A}) * \underline{B}$$

① Translate Shift

② Scale

$$z_{11} \rightarrow z_{11}^N \rightarrow g(z_{11}^N) \rightarrow a_{11} \quad \times$$

$$z_{11} \rightarrow z_{11}^N \rightarrow \underline{z_{11}^{BN}} \rightarrow g(z_{11}^{BN}) \rightarrow a_{11}$$

* Sometimes your NN does not want to Batch Normalization

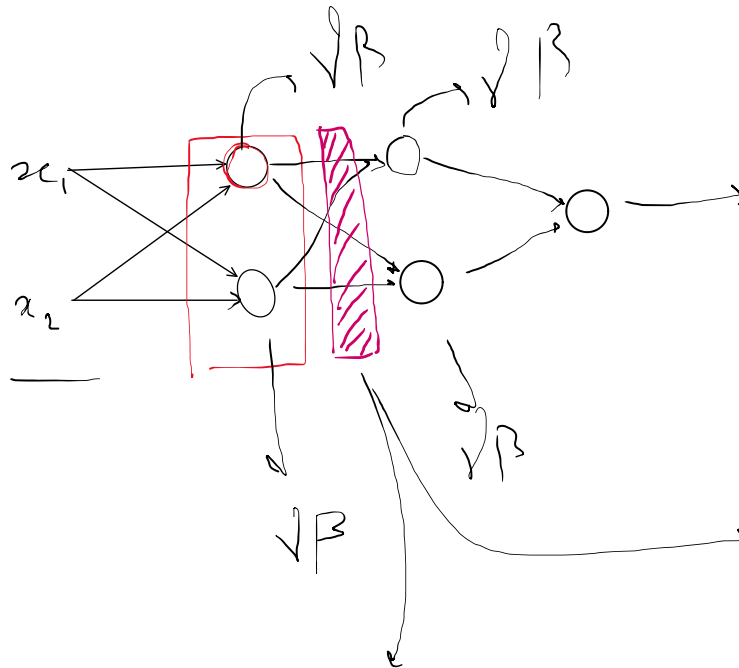
$$z_{11}^{BN} = 1 * z_{11}^N + 0 \quad \leftarrow$$

Scale & Shift $\rightarrow$
$$z_{11}^{BN} = \gamma z_{11}^N + \beta \quad \leftarrow \text{what if it wants}$$

flexibility $\rightarrow$
$$[\gamma, \beta] \leftarrow (\text{trainable parameters})$$

$$\gamma = \gamma - \eta \frac{dL}{d\gamma} \quad | \quad \beta = \beta - \eta \frac{dL}{d\beta}$$

$$\gamma = \gamma - \frac{1}{N} \sum_{i=1}^N \gamma_i$$



4 trainable
params for γ

4 non trainable params
 σ, μ

What happens during Test ?

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[1 guy point] $\rightarrow \boxed{\mu_B \quad \sigma_B} ?$

$\frac{100 \text{ slow}}{\text{batch} = 4} \left\{ \begin{array}{l} 25/\text{epoch} \end{array} \right.$

0 γ_B
0 $\mu'_B \quad \sigma'_B$

E W M A

Exponential Weighted Moving Average

$1 B \rightarrow \mu_{B_1} \quad \sigma_{B_1}$
 $2 B \rightarrow \mu_{B_2} \quad \sigma_{B_2}$
 $\vdots$
 $25 B \rightarrow \mu_{B_{25}} \quad \sigma_{B_{25}}$

$\boxed{\mu'_B \quad \sigma'_B}$

$\rightarrow$ Stable

$\rightarrow$ Factor

$\rightarrow$ Regularization to some extent.